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NFC-Based Vehicle Ignition System

Theme Based Project

Submitted in partial fulfillment of the academic requirement for the award of the degree of

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In

ELECTRONICS AND COMMUNICATION ENGINEERING

By

B.PRANEETH	**1602-23-735-089**
M.SHIVANAND	**1602-23-735-108**
C.VIVEK	**1602-23-735-127**

Under the guidance of
Mr. V. Krishna Mohan
Designation: Associate Professor
E.C.E Department

Department of Electronics and Communication Engineering

Vasavi College of Engineering (Autonomous)

ACCREDITED BY NAAC WITH 'A++' GRADE

IBRAHIMBAGH, HYDERABAD-500031

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Department of Electronics and Communication Engineering

Vasavi College of Engineering (Autonomous)

ACCREDITED BY NAAC WITH 'A++' GRADE

IBRAHIMBAGH, HYDERABAD-500031

Date:

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This is to certify that the Project work titled **"NFC-Based Vehicle Ignition System"** submitted by

B.PRANEETH	**1602-23-735-089**
M.SHIVANAND	**1602-23-735-108**
C.VIVEK	**1602-23-735-127**

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Coordinators
Theme Based Project
E.C.E Department

Head of the Department
Dr. E. SREENIVASA RAO
Professor and HOD
E.C.E Department

External Examiner

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B.PRANEETH	**1602-23-735-089**
M.SHIVANAND	**1602-23-735-108**
C.VIVEK	**1602-23-735-127**

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B.PRANEETH **1602-23-735-089**

M.SHIVANAND **1602-23-735-108**

C.VIVEK **1602-23-735-127**

ABSTRACT

An advanced ignition system for vehicles that uses Near Field Communication (NFC) technology and facial recognition technology to enhance vehicle security. The ignition system consists of a Raspberry Pi 5 platform, an ESP32 platform along with PN532 reader module, and two levels of verification to prevent unauthorized access from being able to start a vehicle.

When a user has an authorized NFC card and taps it on the PN532 reader module, the ESP32 reads the card details and sends them to the Raspberry Pi. The Raspberry Pi compares the card details with the stored information to verify the user's authorization to use the vehicle. If the card details are valid, the Raspberry Pi will turn on the camera and take a picture of the user's face.

The Raspberry Pi then compares the captured image of the user’s face with the images of authorized users stored in the database. If the user’s face matches one of the stored images, the Raspberry Pi will send a signal to the ignition system to allow it to turn on. If the NFC card details were invalid or the user’s face does not match any of the stored images, the ignition system will remain in a locked state, and the ignition will be disabled so that the vehicle cannot be started.

The system enhances the security of the vehicle because it uses an NFC card as well as facial recognition to provide two levels of verification that ensure only authorized users can start the vehicle and reduces the likelihood of unauthorized users being able to start the vehicle.

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CHAPTER – 1

INTRODUCTION

1.1 Introduction:

Regular keys can provide a significant risk if a person loses their key or has their key copied. In order to minimize this risk, we utilize both NFC and facial recognition to access a vehicle. When attempting to start the vehicle, it checks both the user’s facial feature and the card together to ensure that the vehicle is secure. In addition, the ability to use this type of access control can create an interactive smart environment (i.e. using your smartphone to access your vehicle) making it convenient for users.

1.2 Aim:

To design and implement a car ignition system using NFC and face recognition technologies on Raspberry Pi.

1.2 Objectives

- To develop an authentication system based on NFC(Near Field Communication).
- To implement verification using face recognition system.
- To implement and improve vehicle security using the developed system.

1.4 Methodology

1. Understanding NFC,face recognition algorithms and hardware components.
2. Interfacing components along with esp32,raspberry pi5.
3. Writing the required code.
4. Software Verification.
5. Implementation on Raspberry Pi 5.

CHAPTER - 2

LITERATURE REVIEW

One common use of NFC (Near Field Communication) is for short range wireless communication. It allows for rapid and secure exchange of information between a card or smartphone and a reader. Within automotive applications, NFC is primarily used for keyless entry and authentication of only authorized cards and devices. The quick response and ease-of-use associated with NFC make it conducive to applications that require real-time capabilities.

Face recognition is another critical security technology that uses the physical characteristics of an individual (i.e., face) as the means to verify identity via comparison against an individual's stored image(s). Computer vision methods are typically employed to perform this type of verification. OpenCV (Open Source Computer Vision Library), an open-source solution, has become one of the most utilized computer vision libraries to achieve face based identity verification via methods including face detection, feature extraction, and face-matching.

Raspberry Pi has become a popular solution for this type of application because it is an economical and efficient computing device. Raspberry Pi has the potential to process real-time data and provide the ability to connect with a variety of other hardware platforms (i.e., cameras and other peripherals), allowing for the integration of multiple technology platforms into one cohesive system; as well, the inclusion of support for both Python and OpenCV simplifies the development process.

Multiple authentication types have gained popularity over the last few years as a way to increase security. Card based access systems combined with facial recognition systems provide a far superior level of security than single-layered systems, thus decreasing the ability for an unauthorized party to gain access and increasing the overall reliability of the system.

Overall, the combination of NFC technology and facial recognition using an embedded system creates an effective vehicle access system that is smart and secure.

Chapter 3

Process Flow Diagram

3.1 Flow Diagram

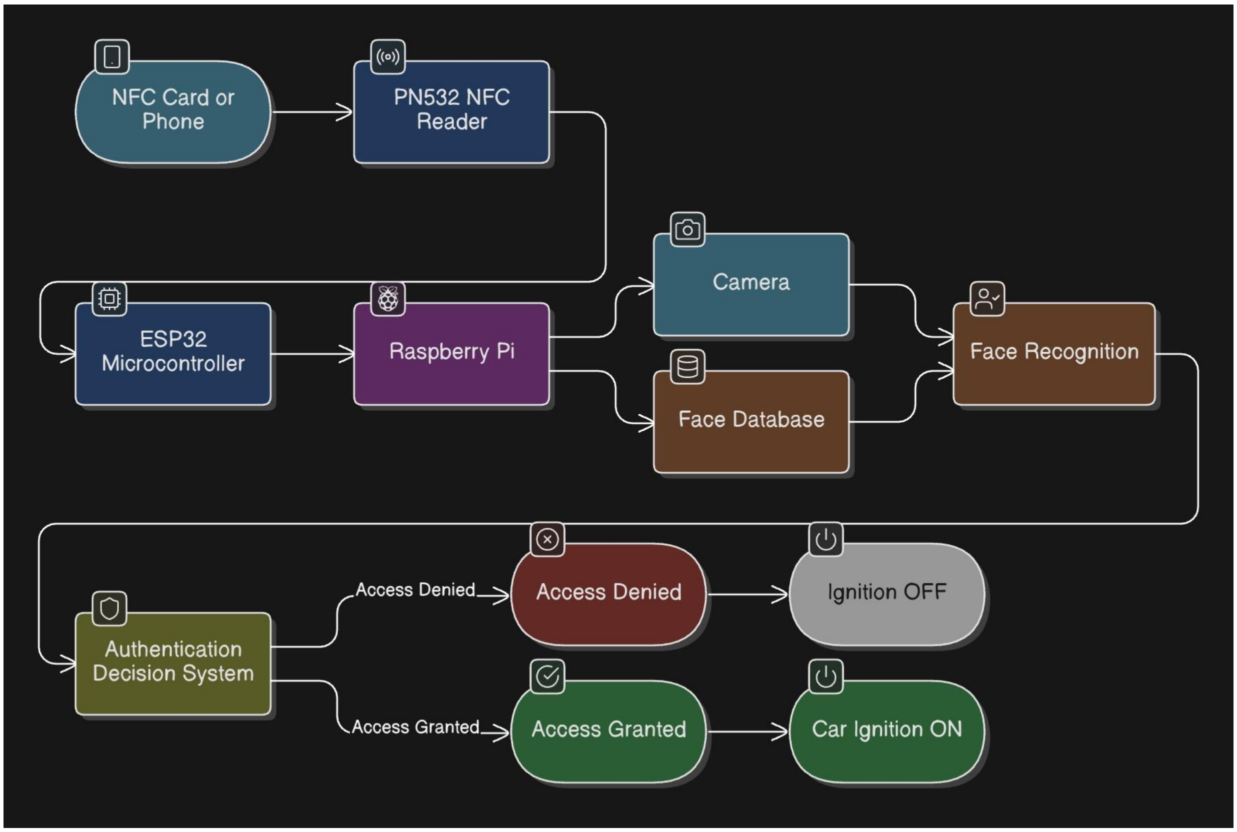


Figure 3.1: Process Flow

3.2 Explanation

- Tapping an NFC card or phone next to the NFC reader starts the system, and the NFC reader scans the card and reads the unique ID of that card from the NFC reader, and this ID will be sent to the ESP32 controller for processing.

- The ESP32 receives the card information and sends it to the Raspberry Pi, where it validates the ID of that card with the database of ID cards stored in memory.

- When the card ID is validated, the Raspberry Pi controls the camera to take a picture of the user's face to complete the verification process. Facial recognition technology will then be used to process the image from the camera and check to see if the face in that image matches any of the faces of the authorized users stored in the database.

- If the face matches, the system has confirmed that the user is authorized. The system checks both the card and face verification process so if someone receives this authorization card, they must also be the correct person.

- If both the card and face are successfully validated as authorized, the Raspberry Pi sends a signal to the ignition control system to activate the relay or motor that allows the engine to start.

- Access is denied to the vehicle if the NFC card is not a valid card or the scanned face does not match the NFC card data. The system keeps the ignition switched OFF, to avoid using the vehicle without proper authorization.

Chapter 4

Project Implementation on Raspberry Pi

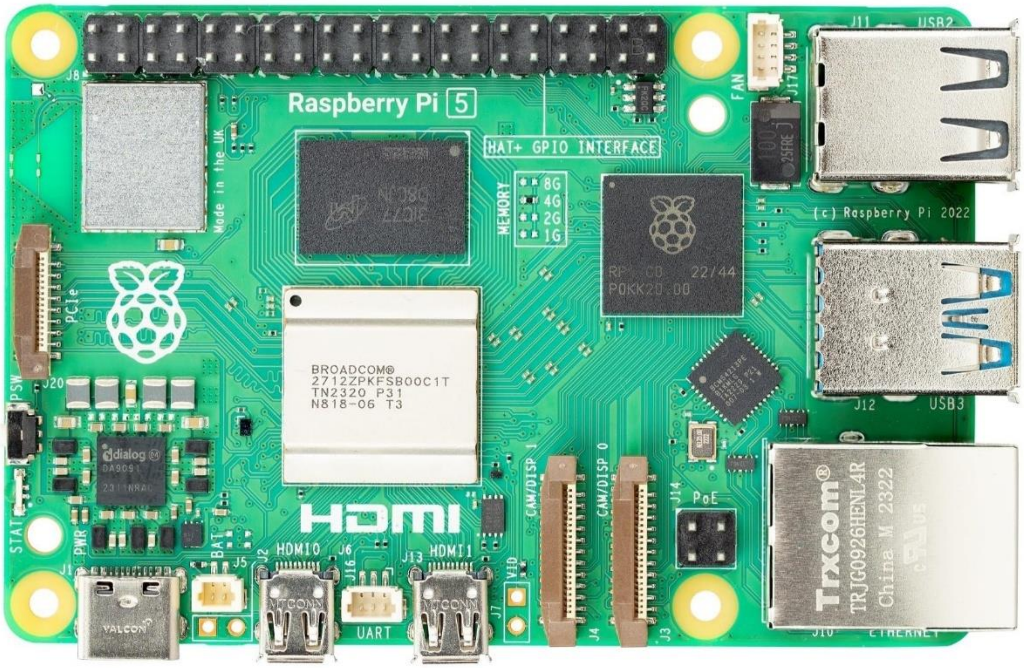


Figure 4.1: Raspberry Pi 5 Single Board Computer

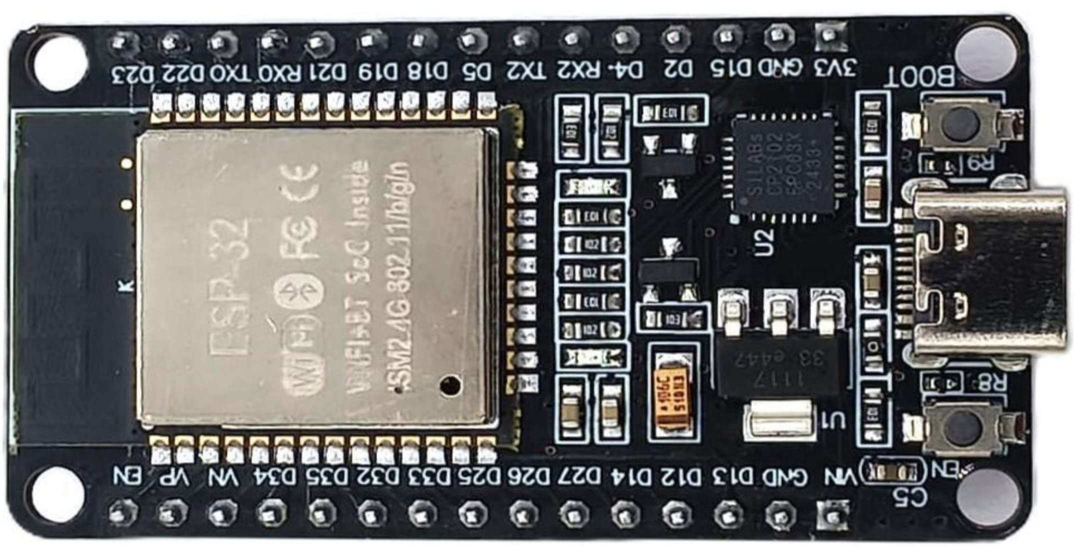


Figure 4.2: ESP32

Steps to Implement:

- **Select the Hardware Board**
  A Raspberry Pi 5 was chosen as the main processing unit, along with a ESP32 microcontroller for communication along with a PN532 module for NFC scanning and webcam for capturing facial data.

- **Configure Hardware Connections**
  Connect both the ESP32 and PN532 modules to the Raspberry Pi using USB-to-UART serial communication. Attach the webcam to the USB port and make sure connections are properly made to avoid communication issues.

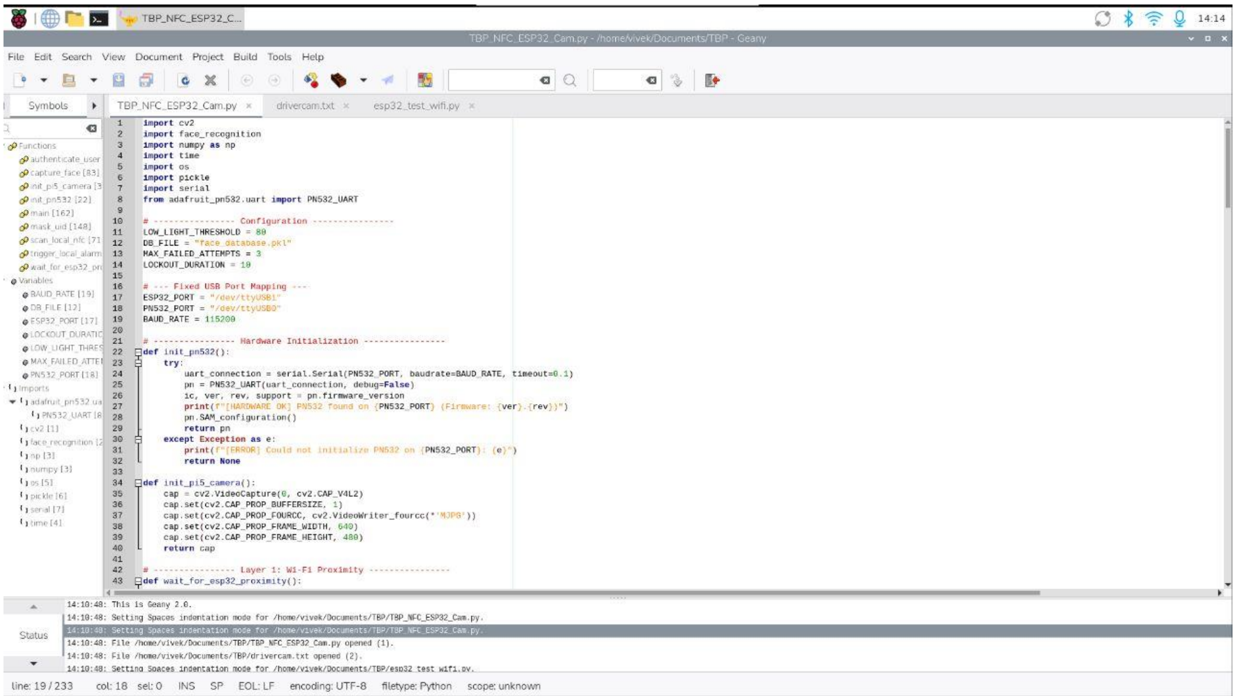
- **Setup The Environment**
  Install all required software on your Raspberry Pi with Raspberry Pi OS to support camera/NFC operation (i.e.: install OpenCV and configure system settings).

- **Validate the hardware mapping**
  Verify from all peripherals to confirm the camera and NFC module are attached and are working properly through the terminal (i.e.: check, through terminal commands, that the ESP32, NFC module, and camera are functioning correctly without issue).

- **Integrate the software architecture**
  Combine all the modules into one central program. Enable seamless interaction between each of the NFC module, facial recognition module and the control logic for continuous operation.

- **Run System Implementation**
  Run the Central Program on the Raspberry Pi. The System will monitor for Inputs, validate an NFC card, compare the facial recognition and display the appropriate outputs.

Code for Implementation of NFC-Vehicle Ignition System:



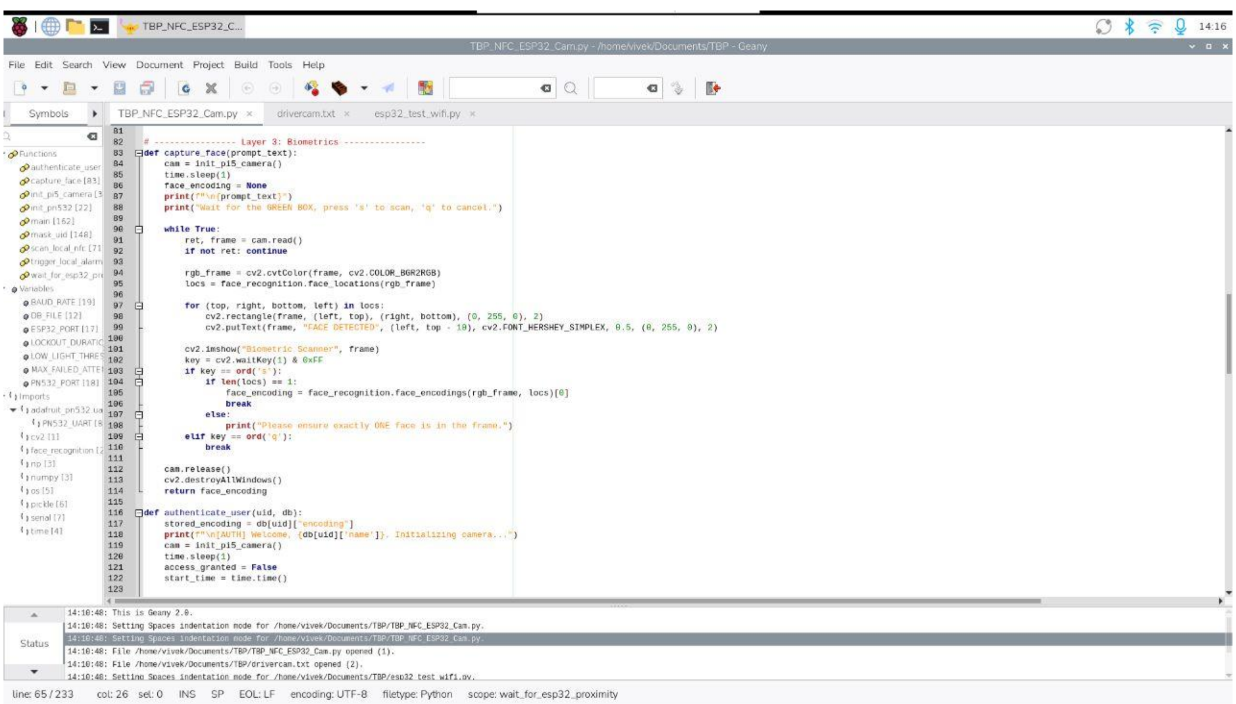
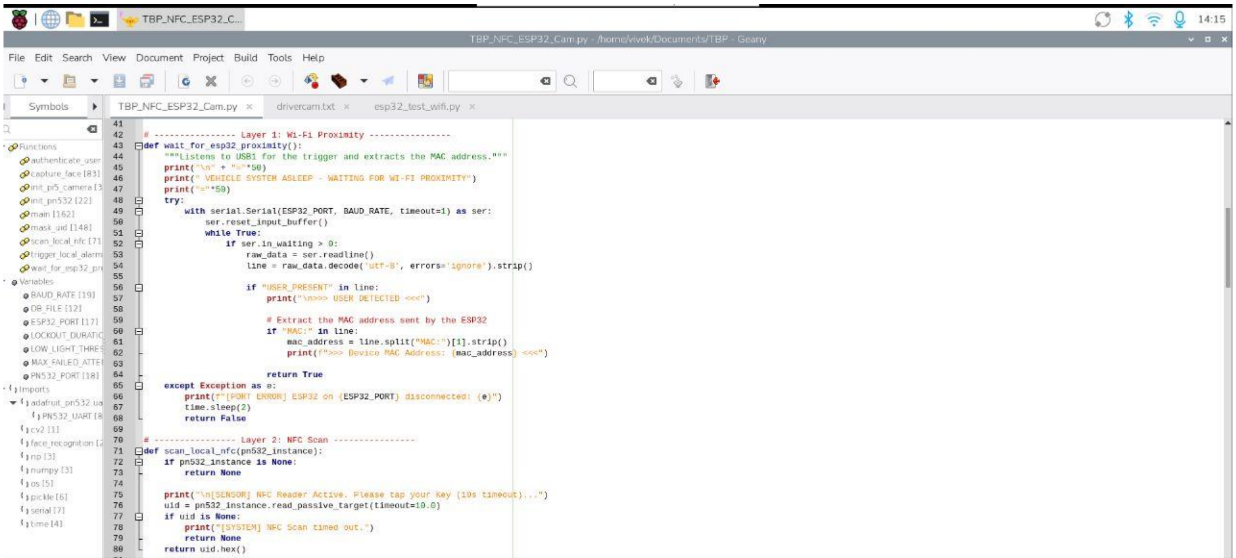


Figure 4.3,4.4:Code Snippets



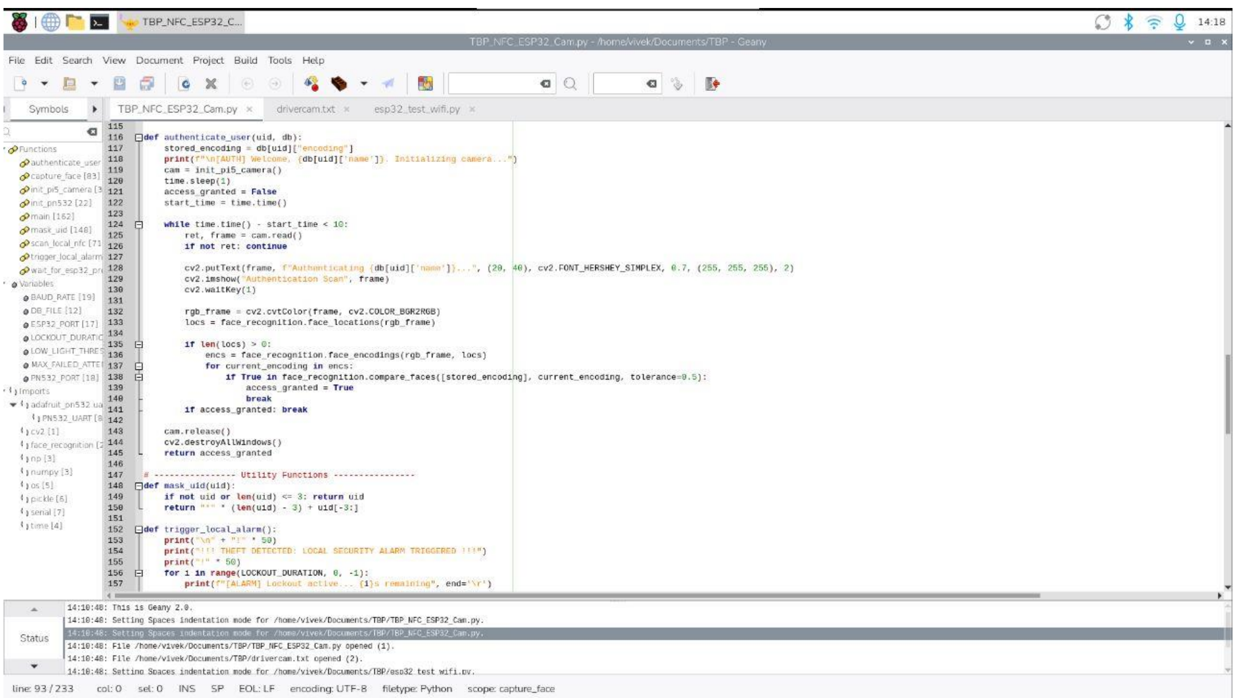
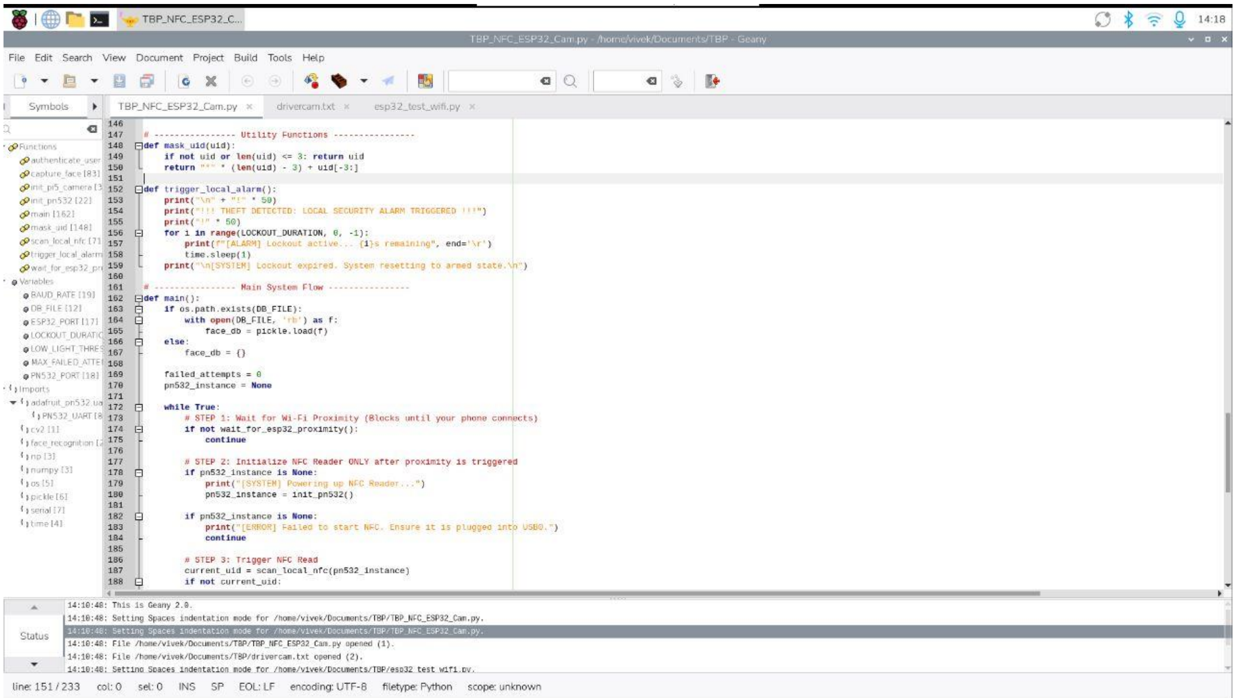


Figure 4.5,4.6:Code Snippet



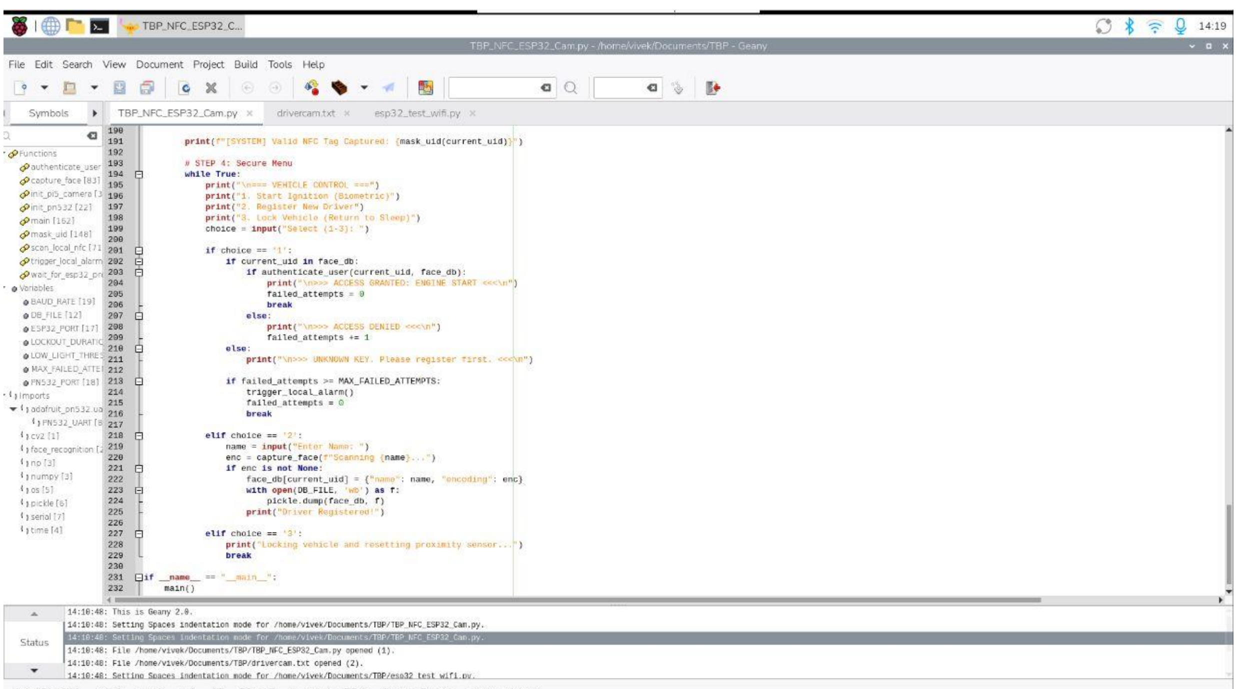
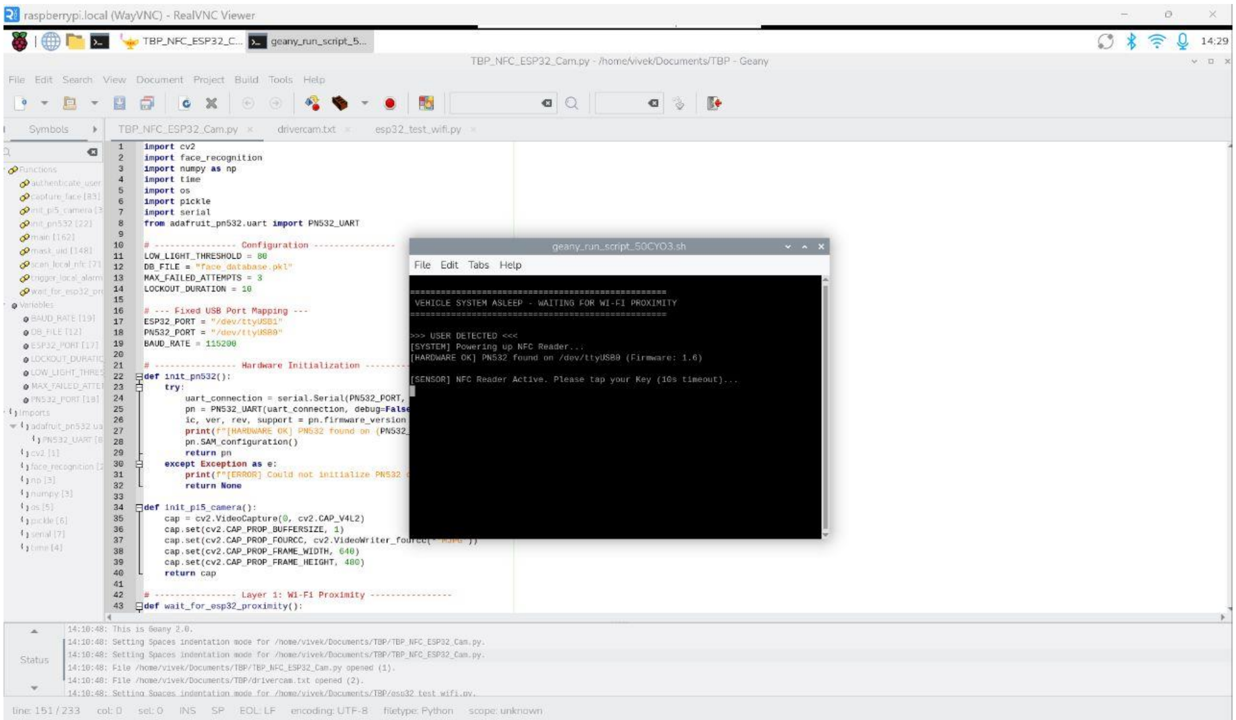


Figure 4.7,4.8:Code Snippet



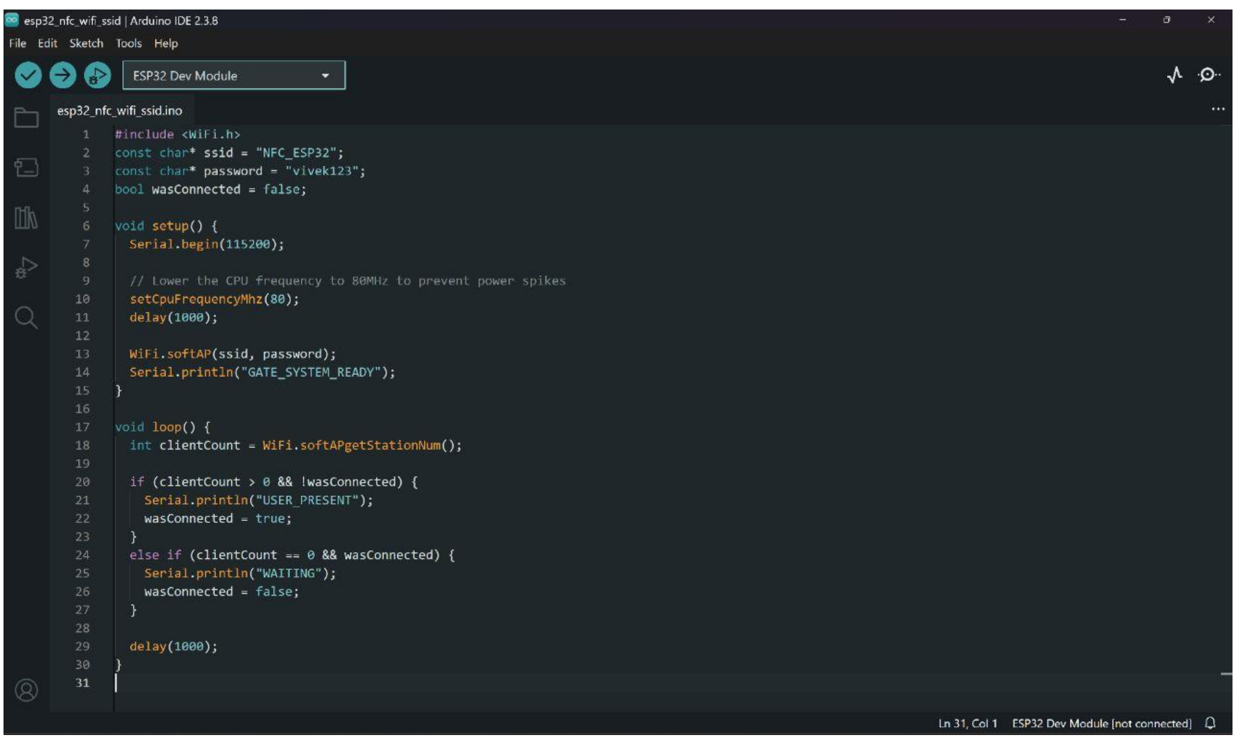


Figure 4.9,4.10:Code Snippet

Code Explanation:

The program primarily utilizes some essential Python libraries that cater to various components of the system. For instance, the cv2 (OpenCV) library is used for accessing the camera, reading frames from the video stream, and showing video with bounding boxes around the faces being detected. The face_recognition library enables face detection and conversion of faces into numeric vectors. The serial library facilitates communication between Raspberry Pi 5 and ESP32, while the PN 532 UART library helps to read data from the NFC chip. The pickle module saves and loads the file of faces database.

Some configuration parameters, such as a name of database file, serial ports, baud rate, as well as limit values like maximum number of failed attempts, are set at the beginning of the script. Those parameters can be adjusted by developers to modify certain aspects of the program behavior without altering its core functionality.

The function **init_pn532()** allows initializing the NFC reader component. First of all, it creates a serial connection and determines whether PN532 reader is correctly connected. In case it is true, the function prints the information about reader firmware and prepares it for scanning. Otherwise, the function returns a message about an error.

This function **init_pi5_camera()** is responsible for initializing the pi5 camera. The function uses OpenCV to initialize the camera for capturing video and sets the camera properties (e.g., frame width and height) to allow for a seamless capture of images for facial recognition.

This function **wait_for_esp32_proximity()** will listen for received serial data from the ESP32 and actively look for a serial message such as "USER PRESENT". This is a signal to the system that a user device is in close proximity, and therefore the system will continue to the next step.

The purpose of the function **scan_local_nfc()** is to use the NFC reader feature to read any NFC card (tags, cards, etc.) that may be presented to the reader. The function will activate the reader and then wait for an NFC card to be tapped on the reader. If an NFC card is detected, the function will return the UID of that card. If no NFC card is detected after a pre-determined time, the function will return nothing.

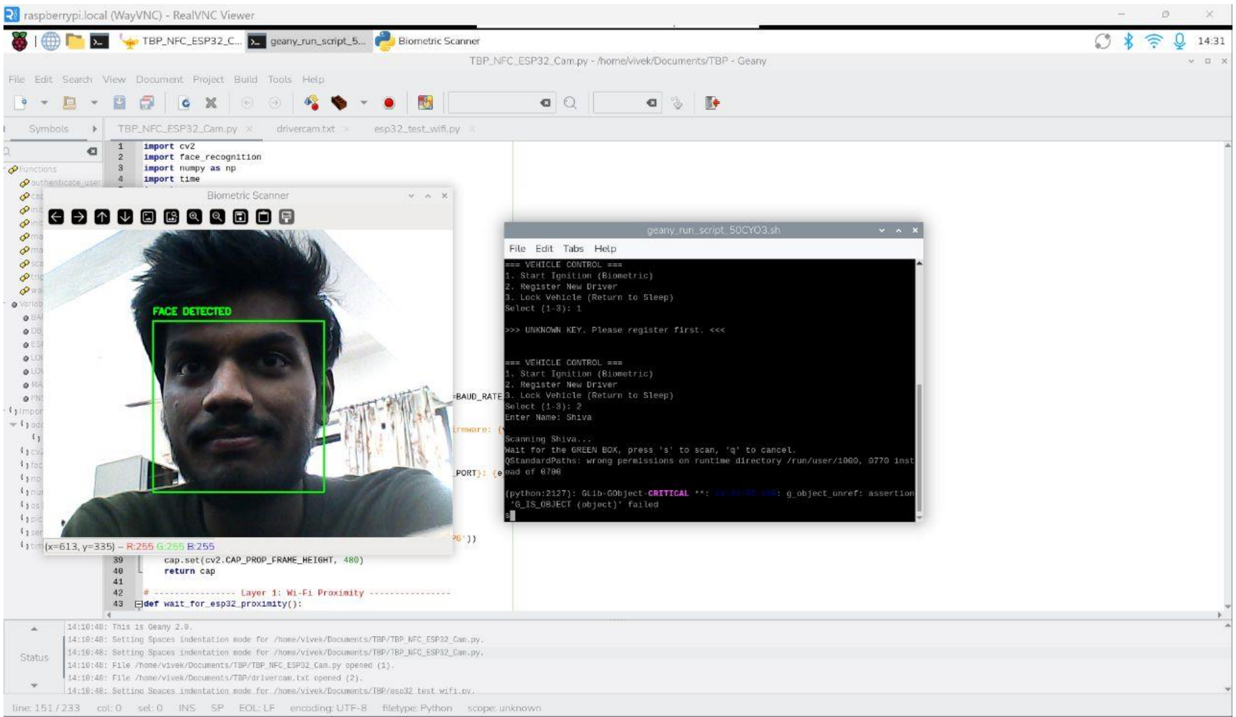
The purpose of the function **capture_face()** is to complete the facial recognition process. The function provides the necessary frames from the camera that will then be analyzed for faces. If a face is detected, a rectangle is drawn around that face and the user confirms the presence of that face. Once one face is confirmed, the function will encode the face and provide that information to later be used for matching.

The function **authenticate_user()** compares the captured face with the face stored in database. It checks if both encodings/features match within a certain tolerance. If a match is found, it returns true causing access to be granted. Otherwise, it returns false.

There are also small helper functions like **mask_uid()** which hide the NFC UID from display, and **trigger_local_alarm()** which handles situations where multiple failed attempts occur. The ***main()*** function controls the overall flow of the program.

Chapter 5

5.1 Results:



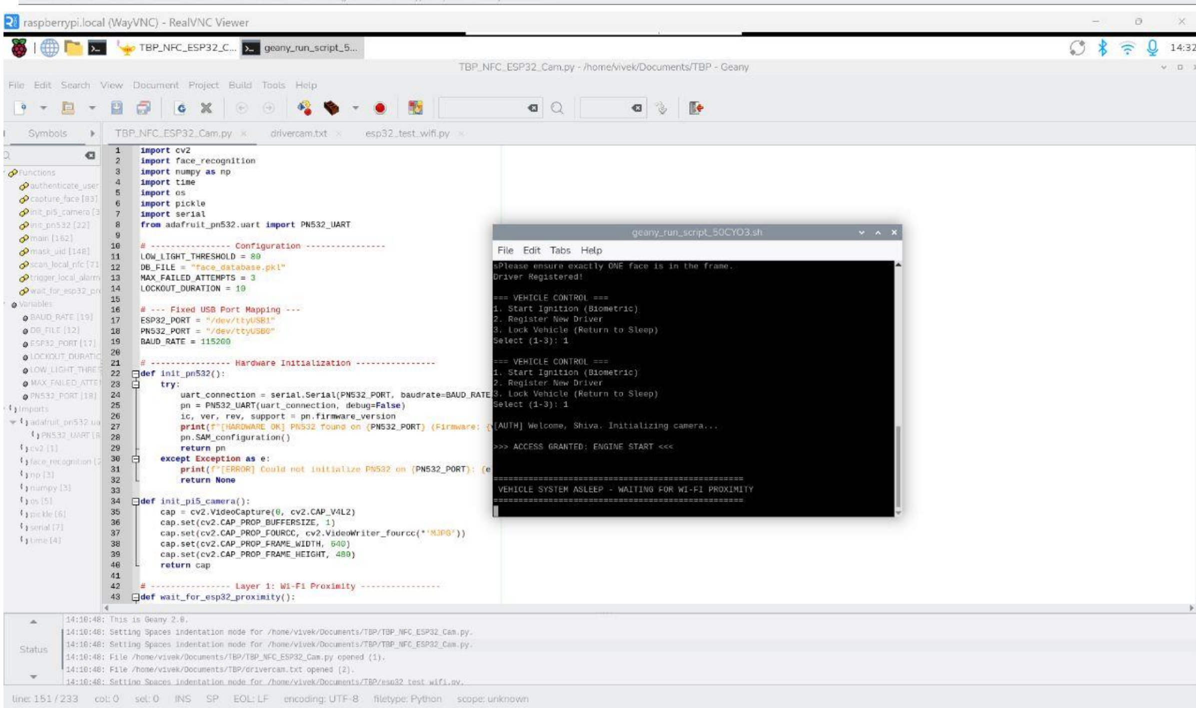


Figure 5.1,5.2 :Output Snippets

- Once the system has been placed into standby mode it will display a menu where options for starting ignition, registering a new user, or locking the vehicle have been presented. This is a clear indication the system can accept input from the user.

- Upon presentation of an unknown user's NFC card, the system recognizes that the NFC card is not a registered NFC card, and it will display a message on the screen such as "Unknown Key – Register for use" to inform the user that only users whose information is stored in the system are permitted to use the vehicle.

- To register as a new user, the system will prompt the user to provide their personal details, while also activating the camera. The camera will take a photo of the user's face when prompted and will record the image of the user's face in the system's database for future authentication processes.

- If the camera was successful in identifying and capturing the user's face, the system will respond by placing a green box on the screen to show "Face Detected", which means the camera and face detection have both been successfully completed.

- Once the new user has been registered, the system will confirm that the user is registered and that the recorded face data has been correctly saved to the system for future access.

- If the registered user returns to the vehicle and presents their registered NFC card and face, the system will check to see that both match before displaying "Access Granted – Engine Start" on the screen. This indicates the user has been successfully authenticated to operate the vehicle.

- After all of the security verification process is complete, the system will return to standby mode waiting for the next user input, or next NFC card scan, indicating that the system is ready for continued operation.

5.2 Conclusion:

The project is a successful demonstration of an ignition system that uses NFC and face recognition techniques to enhance the security of the car. Using both the mentioned technologies allows the system to ensure only registered people can turn on the engine. Therefore, the dual verification system is more reliable than traditional key systems.

It is clear from the used components (Raspberry Pi, ESP32, NFC module, camera), how various embedded technologies can be combined into an easy-to-operate system. As a result, the ignition could be turned on, NFC cards were read, faces were recognized, and so forth.

In general, this project provides the basics of the design of modern security systems based on embedded systems and simple image processing techniques. At the same time, this project shows how such technologies can be implemented in practice to increase security and user comfort. With some additional work, the presented system can become faster and more precise. This system can be further improved by developing various extensions such as an application and a tracking system.

Chapter 6

6.1 Future Scope:

- Add external light sources and/or infrared imaging which will allow for operations in low-light settings.
- Mobile and cloud based solutions will provide solutions for authenticating face images through cloud upload.
- During any unauthorized entry, face images will be uploaded immediately to the cloud.
- Systems installed in automobiles can use enhanced hardware and software for performing in real time.
- Improve algorithms which will increase system accuracy and speed.

6.2 References:

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Chapter 7

SUSTAINABLE DEVELOPMENT GOALS

Sustainable Development Goal 11 – Sustainable Cities and Communities:

The proposed project will help support the development of smarter and more sustainable Cities by improving vehicle access and the overall security of vehicles. Many areas of our world have experienced problems associated with traditional physical key systems, such as when a key is lost, duplicated, or misused. By using NFC and facial recognition, we are able to offer a much safer and better solution for accessing vehicles. This will help to reduce vehicle thefts, in addition to preventing vehicle owners from incurring unnecessary financial losses, which is a given in many of the larger metropolitan cities around the world where vehicle use is high on a daily basis.

This also meets the needs of consumers with a growing move towards digital and contactless technology. Users will be able to access their vehicles with non-physical electronic access methods (using NFC cards or smart phones) rather than physically having to rely on their physical key - which requires less effort than using a traditional physical key and promotes smarter technology within transportation systems that already exist.

As improvements are made on a continual basis to the current non-physical technology being utilized, the ability to connect this technology with mobile applications, GPS based tracking systems, and cloud-based solutions will provide for better established and organized transportation systems. Ultimately, this project demonstrates how digital technology can be utilized practically to support the development of smarter and more sustainable Cities.